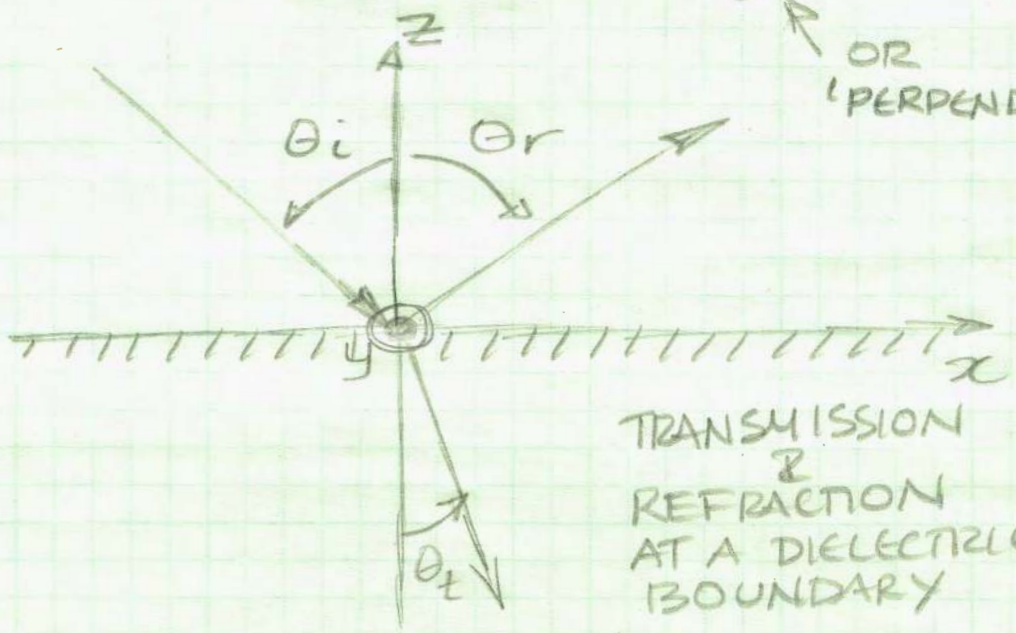
INDEPENDENCE OF TE AND TM FIELDS WHEN THE BOUNDARY IS UNIFORM IN THE y DIRECTION

OR 'PERPENDICULAR'



TRANSMISSION & REFRACTION AT A DIELECTRIC BOUNDARY

- WHEN THE BOUNDARY IS UNIFORM IN THE y DIRECTION, AND THE PLANE OF INCIDENCE LIES IN THE x-z PLANE, THE TE & TM WAVES ARE <u>INDEPENDENT</u> OF EACH OTHER
- THAT IS, A TE INCIDENT WAVE CAN NOT TRANSFORM OR COUPLE TO A TM WAVE UPON REFLECTION OR REFRACTION, AND VICE VERSA!

THIS CORRESPONDS TO THE CASE THAT WE HAVE BEEN STUDYING WHERE THE BOUNDARY IS A PLANE.

CONSIDER

$$\nabla \times E = -j\omega\mu H$$

$$\nabla \times H = j\omega\epsilon E$$

EXPAND IN RECTANGULAR COORDINATES.

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & 0 & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}$$

$$j\omega\mu H_x = \partial E_y/\partial z$$

$$j\omega\mu H_y = \partial E_z/\partial x - \partial E_x/\partial z$$

$$j\omega\mu H_z = -\partial E_y/\partial x$$

$$j\omega\epsilon E_x = -\partial H_y/\partial z$$

$$j\omega\epsilon E_y = \partial H_x/\partial z - \partial H_z/\partial x$$

$$j\omega\epsilon E_z = \partial H_y/\partial x$$

IT'S FAIRLY APPARENT THAT IF $\partial/\partial y = 0$, THE EQUATIONS DIVIDE INTO INDEPENDENT SETS,

$$j\omega\mu H_x = \partial E_y/\partial z$$
$$j\omega\mu H_z = -\partial E_y/\partial x$$
$$j\omega\epsilon E_y = \partial H_x/\partial z - \partial H_z/\partial x$$

TE

$$j\omega\mu H_y = \partial E_z/\partial x - \partial E_x/\partial z$$
$$j\omega\epsilon E_x = -\partial H_y/\partial z$$
$$j\omega\epsilon E_z = \partial H_y/\partial x$$

TM

BECAUSE THE SETS ARE INDEPENDENT, A TE INCIDENT WAVE CANNOT TRANSFORM OR COUPLE TO A TM WAVE UPON REFLECTION OR REFRACTION OR VICE VERSA IF THE SCENARIO IS TWO-DIMENSIONAL, I.E, UNIFORM IN THE y DIRECTION, $\frac{\partial}{\partial y} = 0$

IF, HOWEVER, THE BOUNDARY IS ROUGH OR OTHERWISE VARIES WITH y SO THAT $\frac{\partial}{\partial y} \neq 0$, THEN COUPLING BETWEEN THE TE & TM MODES CAN OCCUR!